INVARIANTS OF A FINITE GROUP ACTION COMMUTE WITH A FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The invariants of a finite group action on a ring are the kernel of the map sending an element to the tuple of its differences with its translates. used at proof lines 32667-32669. Produces the exact sequence whose exactness under base change gives the statement.
- Tensoring an exact sequence with a flat module keeps it exact, so the sequence describing the invariants survives a flat base change. invoked at proof lines 32669-32671. Justifies keeping the sequence exact after tensoring, which is what identifies the invariants of the base change.

Source reference. Original lemma 0BRG: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32635–32635. Cited at proof lines 32676-32677 for the second statement; the exact sequence argument that produces the first statement, on which the second rests, is written out inside the counted span.

Source reference. Original lemma 09EF: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof line 32677 together with the previous reference to finish the second statement; the reduction on which that use depends is inside the counted span.

Lemma 0.2. *Let R be a ring. Let G be a finite group acting on R. Let $R^G \to A$ be a ring map. The map*

$$A \to (A \otimes_{R^G} R)^G$$

is an isomorphism if $R^G \to A$ is flat. In general the map is integral, induces a homeomorphism on spectra, and induces purely inseparable residue field extensions.

Proof. To see the first statement consider the exact sequence $0 \to R^G \to R \to \bigoplus_{\sigma \in G} R$ where the second map sends x to $(\sigma(x) - x)_{\sigma \in G}$. Tensoring with A the sequence remains exact if $R^G \to A$ is flat. Thus A is the G-invariants in $(A \otimes_{R^G} R)^G$.

The second statement follows from Lemma 0BRG and Algebra, Lemma 09EF. □

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